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Cooling Effects of Roof Greening Based on Local Climate Zone Framework and Geographic Detector Analysis

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Abstract

Green roofs have been shown to significantly mitigate urban heat island effects and serve as an important solution for addressing the shortage in urban green spaces. Their capacity to improve the surface thermal environment is influenced by both the morphology of the urban blocks and their landscape patterns. In this study, surface temperatures for four seasons (spring, summer, autumn, and winter) were retrieved, and the morphological types of the central area of Guangzhou were classified using the local climate zone (LCZ) framework. Vector data on green roofs were collected to construct a cooling model and calculate landscape pattern metrics. A geodetector model was employed to analyze the cooling mechanisms of the green roofs. The results indicated that the cooling effect of green roofs is most pronounced in summer, with the highest average cooling intensity of an area reaching 0.63℃. The optimal LCZ type for green roof cooling performance was found to be that of compact low-rise buildings, and compact building areas were identified as more suitable than open building areas for green roof implementation. Furthermore, larger average green roof areas were associated with enhanced cooling effects.

Keywords Green roofs · Urban Heat Island · Local Climate Zone · Geodetector

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Introduction

As the global economy continues to develop, urbanization has accelerated, leading to rapid urban expansion and high population density. This has exacerbated the urban heat island (UHI) effect and has attracted significant research attention (Santamouris, 2015). The Intergovernmental Panel on Climate Change Fifth Assessment Report by Working Group I highlighted that a 0.85 °C rise was observed in the global average surface temperature over the past 30 years, with the rate of warming nearly doubling compared to 30 years ago (Qin & Stocker, 2014). Global climate warming is closely related to the acceleration of urbanization (Shi et al., 2015; Zhong-Wei et al., 2016). As urbanization progresses, permeable natural surfaces such as water, soil, and vegetation are being extensively replaced by impervious surfaces made of asphalt and concrete. Under the same climatic conditions, these surfaces heat faster and reach higher temperatures than do vegetation and other natural surfaces. This is compounded by the heat generated from human activities, which further intensifies the UHI effect (Peng et al., 2013; Zhang et al., 2010). The UHI effect exemplifies the negative impact of urbanization on climate, contributing to issues such as environmental degradation, increased frequency of health problems owing to extreme weather, high carbon emissions, and aggravated air pollution, thus posing challenges to urban sustainability (Hou et al., 2013; Huang & Lu, 2017; Kovats & Hajat, 2008). Therefore, mitigating and addressing the UHI effect is crucial for achieving sustainable urban development.

In recent years, numerous studies have focused on mitigating the UHI effect by using a series of simulations. By analyzing the causes of UHI and conducting case studies and feasibility analyses, various mitigation strategies have been proposed (Amir et al., 2018; Rizwan et al., 2008), such as expanding urban green spaces, enhancing urban wetlands, increasing water surface areas, improving urban building materials, adjusting land-use strategies, and reducing anthropogenic heat emissions. However, owing to the rapid pace of urbanization, many cities face limitations in land-use patterns, shortage of available land, and increasing population density. Therefore, careful consideration of the feasibility of UHI mitigation policies is necessary. Green roofs, as part of urban green spaces, can effectively reduce the surrounding air and rooftop surface temperatures through evapotranspiration and by blocking and reflecting solar radiation. Because of their low cost and the ability to be deployed on a large scale in urban areas, green roofs are considered one of the most suitable methods for mitigating the UHI effect, especially in cities facing land scarcity and constraints on greenspace expansion (Gong & Luo, 2019). The effectiveness of green roofs in mitigating the effects of UHIs has been widely validated. Wu et al. compared simple and garden-style green roofs in Chongqing and demonstrated that green roofs significantly reduce indoor temperatures and increase humidity (Wu et al., 2015). A study conducted in the Baltimore-Washington metropolitan area found that if green roofs were installed on half of the available rooftop surfaces, the surface temperature in the entire region could decrease by 0.83 °C (Li et al., 2014).

Green roof research methods can be categorized into three scales: micro (buildings), meso (neighborhoods), and macro (cities or regions). The mesoscale is particularly important in guiding urban spatial planning and has been the focus of research because of its relevance to practical efforts to mitigate the UHI effect. Mesoscale studies typically select representative neighborhoods based on typical urban environments or idealized urban spatial types (Gong & Luo, 2019). Current research often selects typical urban environments and constructs ENVI-met models to simulate microclimates and assess the effectiveness of green roofs in mitigating the UHI effect (Feng et al., 2022; Jamei et al., 2023; Zheng et al., 2023). However, the diverse and complex morphology of urban neighborhoods in large cities makes it difficult to explore how different neighborhood morphologies influence the cooling effects of green roofs in typical urban settings. Specifically, most mesoscale studies rely on idealized urban models or single-functional zones, lacking analysis of green roof cooling effects across diverse local climate zone (LCZ) types in high-density cities, particularly comparisons between compact and open block morphologies. Furthermore, existing simulations prioritize canopy-layer UHI, with few quantitative studies on direct surface temperature reduction and inadequate integration of roof landscape patterns with urban morphological parameters. Moreover, although the role of green roofs in alleviating the urban heat island effect has been acknowledged in existing studies, attention has largely been confined to the impact of block morphology, whereas the contribution of the roofs' own landscape patterns has received limited consideration.

This study aims to address these gaps by integrating the LCZ framework with geodetector modeling to unravel the cooling mechanisms of green roofs in complex urban morphologies, with a focus on Guangzhou as a representative case of high-density subtropical cities. Specific objectives include: (1) Compare seasonal heat cooling intensity (HCI) of green roofs across six typical LCZ types in central Guangzhou to identify maximum cooling potential during summer heatwaves. (2) Determine the independent and interactive influences of urban morphological factors and green roof landscape metrics on cooling performance within Guangzhou's mixed-use neighborhoods. (3) Identify the LCZ type with optimal cooling efficiency and propose landscape configuration thresholds to guide spatial planning in Guangzhou and analogous high-density cities. The findings of this study are expected to offer valuable guidance for green roof implementation in Guangzhou and contribute to the transition of the city towards a green, low-carbon future.

Materials

Study Area

To investigate the cooling effects of green roofs, a study area of 8 km^2 was selected within the central urban region of Guangzhou, Guangdong Province, China, based on high-resolution imagery. This selection was driven by Guangzhou's representativeness as a typical Chinese city with a "hot summer and mild winter" climate, where two key factors highlight its suitability: first, the densely populated central commercial and residential districts have led to a persistent UHI effect in recent years, creating an urgent need for urban cooling solutions(Gao et al., 2021); second, most buildings in Guangzhou feature flat roofs, providing substantial potential for green roof development, establishing it as an ideal macro-scale context for exploring green roof efficacy.The selected area exhibits diverse urban block morphologies and ample green-roof installations (Fig. 1). As the core of Guangzhou's urban landscape, The study area integrates diverse functional zones, including commercial districts, residential buildings, schools, hospitals, and parks, mirroring the city's typical urban complexity. It is densely populated with significant green roof development, directly embodying the urban conditions where cooling effects are most observable. Additionally, the area exhibits diverse urban block morphologies and complete LCZ types, allowing analysis of cooling effect variations across heterogeneous environments, while ample green roof installations provide sufficient samples for statistical reliability. This combination of Guangzhou's macro-scale representativeness and the central urban area's functional diversity and practical relevance makes the 8 km^2 region a representative and practical case study for investigating neighborhood-scale green roof cooling effects.

Data Source and Processing

The multisource data used in this study are listed in Table 1. The vector data included (1) land use and road network data for Guangzhou, which were used to divide the study area into urban blocks, serving as the units for subsequent analysis and (2) building data (including building height), which were utilized to calculate various urban morphological structure factors. The remote sensing data included (1) 2020 Google imagery with a spatial resolution of 0.54 m, which was used to identify and vectorize green roofs and (2) 10 scenes of Landsat 8 imagery covering the four seasons, which were used for land surface temperature retrieval. For spring, imagery was selected from February 12, 2018, February 20, 2021, and April 4, 2022; for

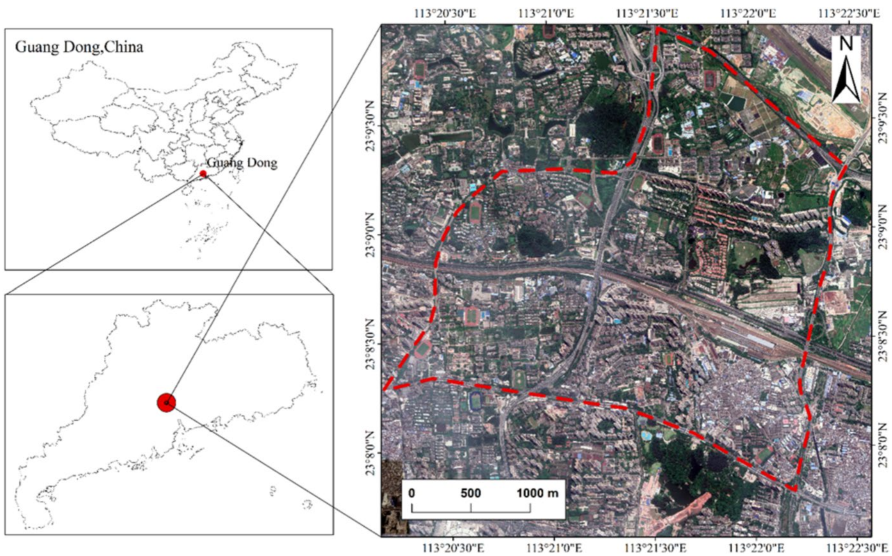


Fig. 1 Location of the Study Area (Guangzhou, Guangdong, China)

Table 1 Data sources used in this study

Type	Resolution	Date	Data Source	Description
Land-use data	-	-	-	Used for research unit division
Road data	-	2019	Open Street Map	
Building data	-	2018	91Weitu Assistance	Used for LCZ classification
Google image	0.54 m	2020	91Weitu Assistance	For the extraction of green roof data
Landsat 8 image	30 m	2018/2/2,2018/7/22,2019/11/14, 2020/2/18,2021/2/20,2022/4/4, 2022/9/3,2022/10/21,2022/12/24, 2023/6/2	USGS	For land surface temperature retrieval

summer, from July 22, 2018, September 3, 2022, and June 2, 2023; for autumn, from November 14, 2019, and October 21, 2022; and for winter, from February 18, 2020, and December 24, 2022.

Data Source of Green Roofs and Block Morphology Classification

Green roof data were collected through the manual interpretation of high-resolution images. Green roofs on the buildings within the study area were vectorized based on a certain scale and continuity. Owing to the inherent height of buildings, there is a positional deviation between the geographical location of green roofs in high-resolution images and their actual geographical location, which could cause errors in subsequent analyses. Therefore, the geographical locations of green roofs must be corrected to the corresponding building foundation locations.

The study area was divided into basic block units using land-use data for Guangzhou, supplemented by main roads. For plots with significant morphological differences within the same unit, further divisions were made based on the actual plots, resulting in 71 block units for analysis.

To explore the relationship between the different morphological characteristics of the blocks and the cooling effects of green roofs, this study applied the local climate zone (LCZ) framework to classify the thermal environment based on surface morphology. The LCZ is a well-developed classification system based on urban climatology that divides zones according to the capacity of the underlying surface to respond to the thermal environment (Stewart & Oke, 2012). Considering the feasibility of data acquisition and quantitative analysis, vector data of building contours (with floor counts) in Guangzhou were obtained, and four urban morphological structural factors were selected for classification: building height (BH), building density (BD), permeable surface fraction (PSF), and sky view factor (SVF) (Benjamin et al., 2015; Zheng et al., 2018). Building height refers to the average height of buildings within a block, building density refers to the ratio of the total building footprint area to the total block area, permeable surface fraction refers to the ratio of the total permeable surface area to the total block area, and the sky view factor is a morphological parameter indicating the extent to which the sky is obstructed by buildings. The formulae for calculating each urban morphological structural factor are as follows:

$$BH = \overline{H}_{build} \tag{1}$$

$$BD = S_{build}/S_A \tag{2}$$

$$PSF = S_P/S_A \tag{3}$$

$$SVF = 1 - \frac{\sum_{i=1}^{n} \sin\gamma_i}{n} \tag{4}$$

where $\overline{H}_{build}$ is the average building height within the block, S_{build} is the total building footprint area within the block, S_A is the total block area, S_P is the total permeable surface area within the block, γ_i is the influence of the terrain height angle for the i-th azimuth angle, and n is the number of azimuth angles calculated.

Based on the LCZ surface attribute classification table proposed by Stewart and Oke (Stewart & Oke, 2012) (Table 2), the study area was classified into 8 building LCZ types (Fig. 2). In the preliminary zoning results, for plots with unclear zoning boundaries or ambiguous attribute characteristics, visuall verification was conducted using Google images. Additionally, the LCZ classification results were further validated through field surveys, during which on-site photos were collected. Field validation results indicated that 85% of the classified LCZ types showed consistency with the actual urban morphology.

Land Surface Temperature Retrieval

Surface temperature is a common indicator for characterizing UHI conditions. Common surface temperature retrieval algorithms include single-band algorithms, dual-band algorithms, and multi-band algorithms (Zhao et al., 2017). Owing to the limited availability of remote sensing images with cloud coverage below 5% in the study area, images with low cloud coverage from recent years (Table 1) for each season were selected to retrieve surface temperatures, ensuring improved seasonal representativeness. The average values of the retrieved surface temperatures were used to represent the surface temperatures for each season. Based on the Landsat 8 TIRS dataset, a radiative transfer equation was used to retrieve surface temperatures for spring, summer, autumn, and winter in Guangzhou.

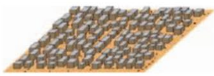
Main Steps:

Step 1: Calculation of Land Surface Emissivity.

Land surface emissivity is a crucial variable for retrieving surface temperatures and is closely related to surface roughness. The normalized difference vegetation index threshold method proposed by Sobrino (Raissouni & Sobrino, 2000) was used in the present study. To obtain more accurate land surface emissivity data, the method proposed by Qin Z. was used, which involves dividing the surface into water bodies, natural surfaces, and urban areas, and calculating the land surface emissivity for each type of surface (Qin et al., 2004). The formulae are as follows:

$$\varepsilon_{\mathrm{water}} = 995 \tag{5}$$

Table 2 Classification of Surface Characteristics of Local Climate Zones (Stewart & Oke, 2012)

Built types	Definition	Values of properties
LCZ1 Compact high-rise 	Dense mix of tall buildings over ten stories. Few or no trees. Land cover primarily impervious surfaces.	BH > 25 m 40% < BD ≤ 60% PSF <10% 0.2 < SVF ≤ 0.4
LCZ2 Compact mid-rise 	Dense mix of mid-rise buildings (3-9 stories). Few or no trees. Land cover primarily impervious surfaces.	10 m < BH ≤ 25 m 40% < BD ≤ 70% PSF < 20% 0.3 < SVF ≤ 0.6
LCZ3 Compact low-rise 	Dense mix of low-rise buildings (1-3 stories). Few or no trees. Land cover primarily impervious surfaces.	3 m < BH ≤ 10 m 40% < BD ≤ 70% PSF < 30% 0.2 < SVF ≤ 0.6
LCZ4 Open high-rise 	Open arrangement of tall buildings (over 10 stories); high vegetation coverage, primarily pervious surfaces.	BH > 25 m 20% < BD ≤ 40% 30% < PSF ≤ 40% 0.5 < SVF ≤ 0.7
LCZ5 Open mid-rise 	Open arrangement of mid-rise buildings (3-9 stories); high vegetation coverage, primarily pervious surfaces.	10 m < BH ≤ 25 m 20% < BD ≤ 40% 20% < PSF ≤ 40% 0.5 < SVF ≤ 0.8
LCZ6 Open low-rise 	Open arrangement of low-rise buildings (1-3 stories); high vegetation coverage, primarily pervious surfaces.	3 m < BH ≤ 10 m 20% < BD ≤ 40% 30% < PSF ≤ 60% 0.6 < SVF ≤ 0.9
LCZ7 Lightweight low-rise 	High-density single-story buildings, few or no trees, ground cover mostly hard soil.	2 m < BH ≤ 4 m 60% < BD ≤ 90% PSF < 30% 0.2 < SVF ≤ 0.5
LCZ8 Large low-rise 	Dense distribution of large low-rise buildings (1-3 stories); little to no vegetation, primarily impervious surfaces.	3 m < BH ≤ 10 m 30% < BD ≤ 50% PSF < 20% SVF > 0.7

BH: Average building height.

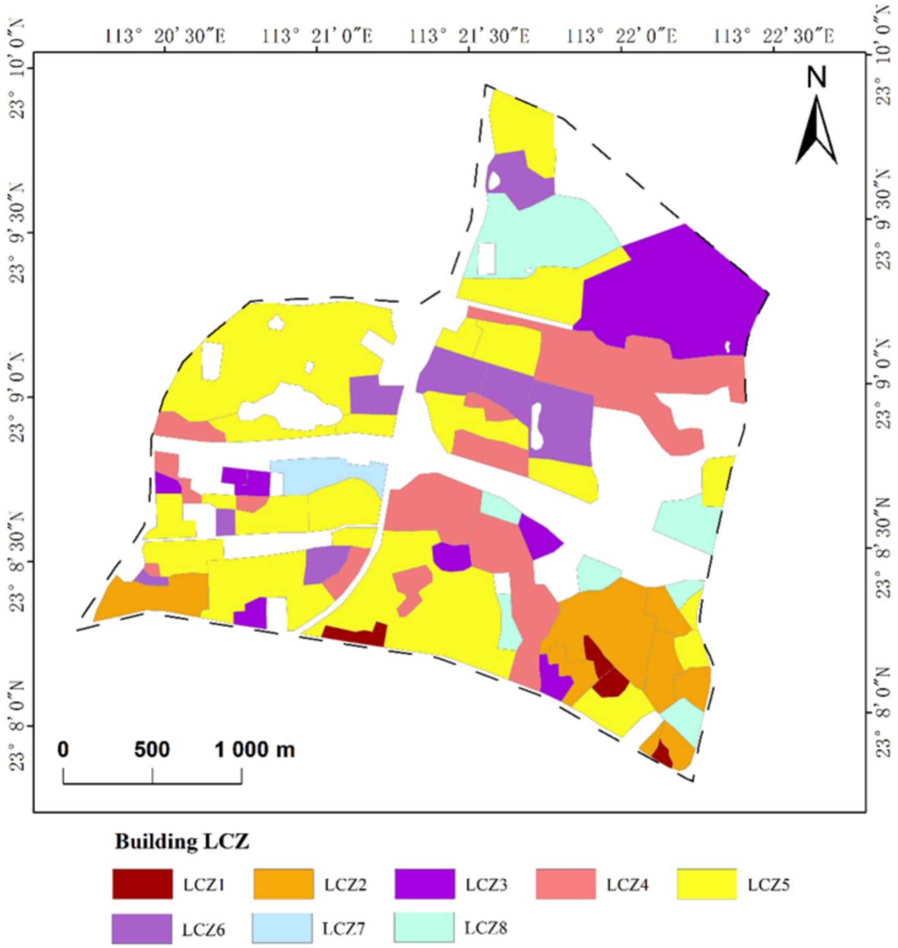


Fig. 2 Classification Results of Urban Building Local Climate Zones (LCZs)

$$\varepsilon_{surface} = 0.9625 + 0.0614Pv - 0.0461Pv^2 \tag{6}$$

$$\varepsilon_{building} = 0.9589 + 0.086Pv - 0.0671Pv^2 \tag{7}$$

where Pv is the vegetation coverage. The calculation formula for Pv is as follows:

$$Pv = \frac{NDVI - NDVI_{soil}}{NDVI_{veg} - NDVI_{soil}} \tag{8}$$

where NDVI is the normalized difference vegetation index, $NDVI_{soil}$ is the NDVI value of completely bare soil or non-vegetated surfaces, and $NDVI_{veg}$ is the NDVI value of pixels completely covered by vegetation, that is pure vegetation pixels.

Step 2: Calculation of Blackbody Radiance

$$B\left(T_s\right)=\frac{\left[L_\lambda-\mathrm{L}\uparrow-\tau\left(1-\varepsilon\right)\mathrm{L}\uparrow\right]}{\tau\varepsilon} \tag{9}$$

where T_s is the actual surface temperature (K), $B(T_s)$ is the blackbody radiance, L_λ is the thermal infrared radiance received by the satellite sensor, L↑ is the upward atmospheric radiance, L↓ is the downward atmospheric radiance, τ is the atmospheric transmittance in the thermal infrared band, and ε is the land surface emissivity.

Step 3: Calculation of Surface Temperature.

Using the Planck function, the formula for calculating T_s is obtained as follows:

$$T_s=K_2/\ln\left(\frac{K_1}{B\left(T_s\right)}+1\right) \tag{10}$$

where $K_1=774.89$ W/(m^2·μm·sr) and $K_2=1321.08$ K.

The average surface temperature data for each season were obtained and masked to generate surface temperature maps for spring, summer, autumn, and winter in the study area (Fig. 3).

Acquisition of Green Roof Landscape Pattern Factors

Green roofs can lower surface temperatures by blocking solar radiation, reducing solar energy that directly reaches the ground, and releasing water vapor into the atmosphere through transpiration, which indirectly cools the surrounding surface. Therefore, understanding how and how many green roofs are planted is crucial for understanding the cooling mechanism and implementing green roofs (Jiang et al., 2018). The landscape patterns of green roofs describe their spatial distribution and layout, quantify the construction of green roofs, and provide an analytical perspective for the subsequent exploration of cooling mechanisms.

The landscape characteristics of the green roofs were calculated using the landscape ecology software Fragstats 4.2 (https://fragstats.org/). Various metrics were selected from categories such as area, shape, aggregation, edge, density, and proximity. After correlation analysis and selection of important factors, seven landscape pattern factors were selected: average patch area (Area), class area (CA), patch density (PD), shape index (Shape), mean nearest neighbor (MNN) distance, total edge (TE) length, and cohesion index. In this study, the Area represents the average area of all green roofs in the region, CA represents the total area of all green roofs, PD represents the ratio of the number of green roofs to their total area, TE length represents the total perimeter of all green roofs, MNN distance represents the average distance between each green roof and its nearest neighbor, Shape represents the deviation of green roofs from a circular shape, and cohesion index represents the degree of aggregation and dispersion of green roofs. The formulae for the shape and cohesion indices are as follows:

$$I_{shape}=\frac{P}{2\sqrt{\pi A}} \tag{11}$$

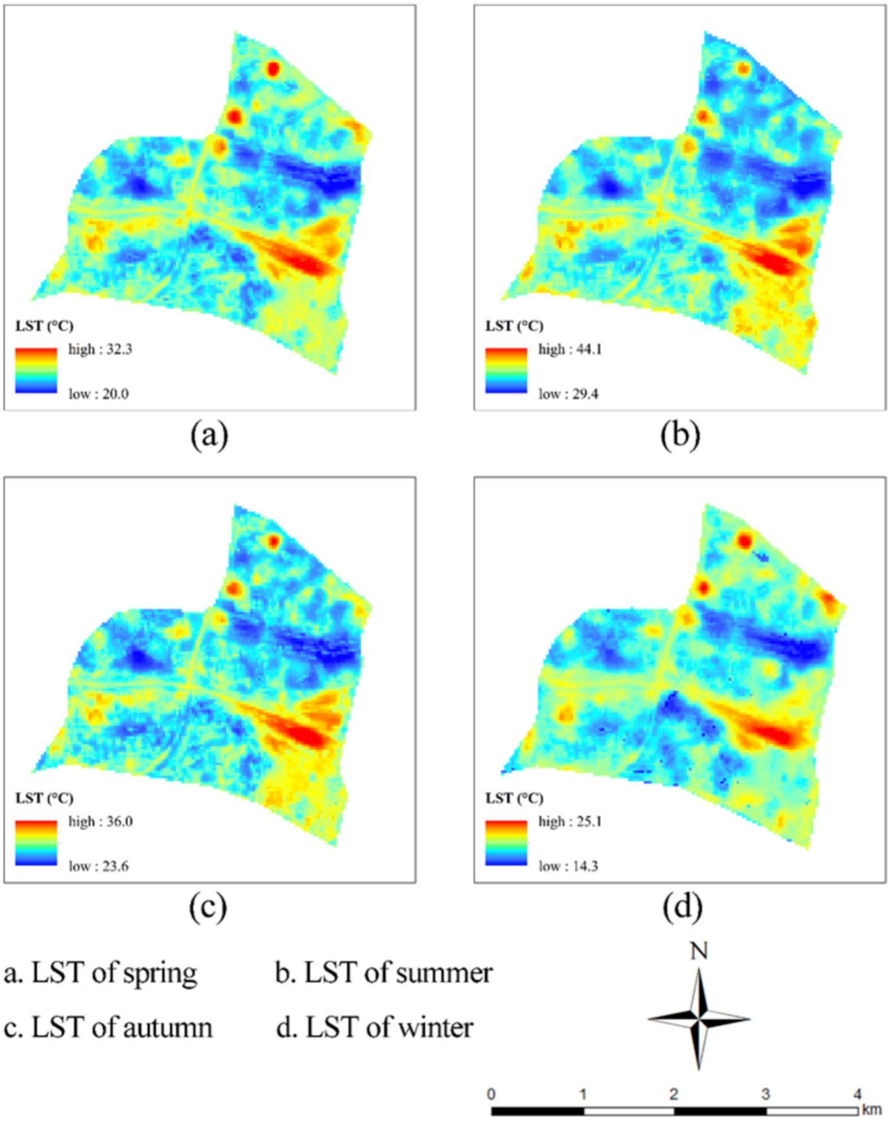


Fig. 3 Land Surface Temperature (LST) Results for Spring, Summer, Autumn, and Winter

$$I_{Cohesion} = \left[1 - \frac{\sum_{j=1}^{m} P_{ij}}{\sum_{j=1}^{m} P_{ij}\sqrt{a_{ij}}}\right]\left[1 - \frac{1}{\sqrt{A}}\right]^{-1} \times 100 \tag{12}$$

where, P is the perimeter of the green roof, A is the area of the green roof, a_{ij} is the area of the j-th patch in the i-th class, and P_{ij} is the perimeter of the j-th patch in the i-th class.

Methods

Green Roof Cooling Intensity

To explore the cooling effect of green roofs on rooftop surface temperatures in different LCZs, a green roof cooling intensity model (HCI_G) was constructed as follows:

$$HCI_G = T_I - T_G \tag{13}$$

where HCI_G represents the cooling intensity of the green roofs, T_I denotes the average surface temperature of a particular LCZ, and T_G denotes the average surface temperature of the green roofs in the LCZ. This model visually demonstrated the cooling effect of green roofs by calculating the temperature difference between the green roofs and the surrounding area. The higher the cooling intensity, the better is the cooling effect of the green roofs in that area.

Statistical analysis revealed that six out of the evaluated eight LCZ types in the study area had green roof constructions: compact mid-rise buildings (LCZ2), compact low-rise buildings (LCZ3), open high-rise buildings (LCZ4), open mid-rise buildings (LCZ5), lightweight low-rise buildings (LCZ7), and large low-rise buildings (LCZ8), totaling 41 block units. The cooling intensity of the green roofs in the LCZs was calculated to determine the cooling effect.

Geodetector

Geodetector models are statistical tools used to explore spatial differentiation phenomena and their driving mechanisms (Wang & Xu, 2017). The core concept is to use spatial heterogeneity to analyze the consistency of the distribution patterns of the independent variable X and dependent variable Y, thereby detecting the explanatory power of the independent variable on the dependent variable. The geodetector model includes four modules: factor, ecology, interaction, and risk. The geodetector analysis process was implemented in an R environment (Version 4.3, https://www.r-project.org/), using green roof cooling intensity (HCI) as the numerical dependent variable Y, and local climate zone types (LCZtype) and landscape pattern indices (Area, CA, PD, Shape, MNN, TE, Cohesion) as categorical independent variables. Factor, interaction, and risk detectors were used to comprehensively explore the cooling mechanism of the green roofs. In geodetector analysis, the input independent variable data must be categorical; therefore, discretization of the independent variable data is required. This discretization process uses the GD package in R 4.3, which can achieve optimal discretization of continuous variables using methods such as equal intervals, natural breaks, quantiles, geometric intervals, and standard deviations. This process is known as optimal discretization. The optimal discretization of landscape pattern indices and complete geodetector analysis were implemented by constructing models in R 4.3.

Results

Green Roof Cooling Intensity

The HCI was statistically analyzed according to the LCZs, as shown in Fig. 4. The HCI values of the six LCZ types in the study area were analyzed to examine the relationship between the HCI and LCZ type. Overall, the green roofs exhibited a cooling effect on the surface temperature of the area. The LCZ types with the best green-roof cooling effects were LCZ3 and LCZ2. In LCZ3, green roofs had a marked cooling effect in all seasons, with the best cooling effect in summer, ranging from −0.18 to 1.18 and averaging 0.63°C. Green roofs in LCZ2 also had a significant cooling effect in all seasons, with the best cooling effect in summer, ranging from −0.18 to 0.92 and averaging 0.38°C. The HCI values in LCZ4 and LCZ5 were close to 0°C in all seasons, indicating no significant cooling effect; in LCZ7 and LCZ8, the HCI values mostly ranged from −0.5 to 0, indicating unsatisfactory cooling effects.

The superior cooling performance of LCZ3 and LCZ2 highlights the critical role of urban morphological compactness in enhancing green roof efficiency. Compact low-rise (LCZ3) and mid-rise (LCZ2) areas typically have higher building density and lower sky view factor, which reduce heat dissipation loss and amplify the localized cooling effect of green roofs through evapotranspiration and shading.

Among spring, summer, autumn, and winter, summer had the highest average surface temperature and was the best season for green roof cooling. Among seasons, LCZ2 and LCZ3 exhibited higher green roof cooling intensities in summer than in other seasons, indicating that green roofs in specific LCZs can achieve the best cooling effect in seasons when cooling is most needed.

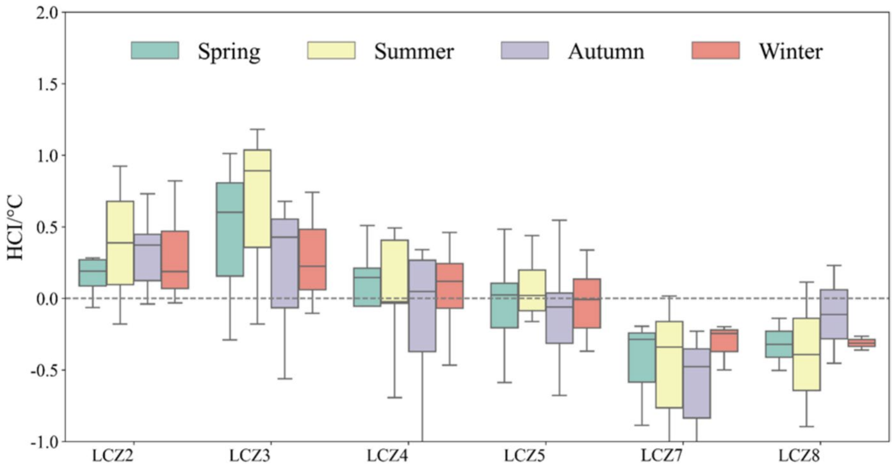


Fig. 4 Statistical Results of Roof Greening Cooling Intensity for Different Local Climate Zones (LCZs) in Spring, Summer, Autumn, and Winter

Green Roof Cooling Mechanism

The factor module in geodetector can determine the influence of LCZ types and green roof landscape pattern factors on the HCI in different seasons, as measured by the q-value. The higher the q-value, the greater is the influence of the factor on the spatial variation in HCI. The detection results (Fig. 5) show that the q-values of the eight factors in each season were all significant. The eight factors were ranked by the q-value in each season to analyze their influence on the spatial variation in HCI. In spring, the order of influence, from the largest to smallest, was Area (0.323) > CA (0.160) > LCZtype (0.159) > Cohesion (0.154) > PD (0.142) > Shape (0.140) > MNN (0.118) > TE (0.109). In summer, the order of influence was Area (0.256) > LCZtype (0.179) > Shape (0.162) > PD (0.156) > Cohesion (0.150) > TE (0.143) > MNN (0.127) > CA (0.092). In autumn, the order of influence was Area (0.211) > TE (0.160) > LCZtype (0.146) > Cohesion (0.141) > Shape (0.128) > MNN (0.119) > PD (0.070) > CA (0.060). In winter, the order of influence was Area (0.223) > Cohesion (0.169) > CA (0.153) > PD (0.125) > LCZtype (0.118) > TE (0.110) > Shape (0.097) > MNN (0.078). The q-value for Area was the highest across all seasons, and among factors, LCZtype ranked in the top four in spring, summer, and autumn. This indicates that Area and LCZtype are the main factors influencing the spatial variation in HCI, whereas shape and MNN have a lesser impact.

Interaction detection was conducted for two main factors: area and LCZtype. When two dominant factors are combined, an interaction pair is formed. The interaction detection results (Table 3) indicated that the q-values of this interaction pair were greater than those of the individual factors. This implies that the interaction

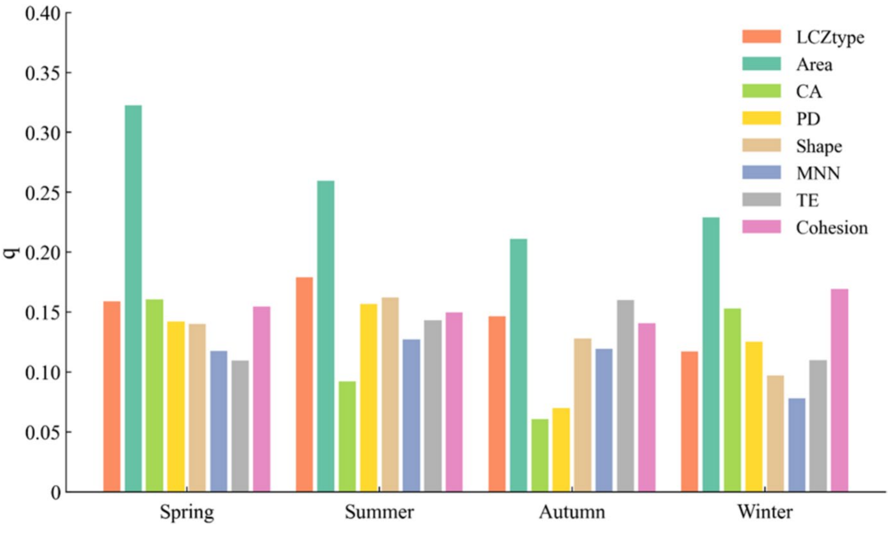


Fig. 5 Detection Results of Factors Affecting Roof Greening Cooling Intensity. (LCZtype: Built types; Area: average area of all green roofs in the region; CA: total area of all green roofs; PD: ratio of the number of green roofs to their total area; Shape: deviation of green roofs from a circular shape; MNN: the average distance between each green roof and its nearest neighbor; TE: the total perimeter of all green roofs; Cohesion: the degree of aggregation and dispersion of green roofs)

Table 3 Interaction Detection Results

Season	q statistic
	Area ∩ LCZtype
Spring	0.596
Summer	0.643
Autumn	0.603
Winter	0.516

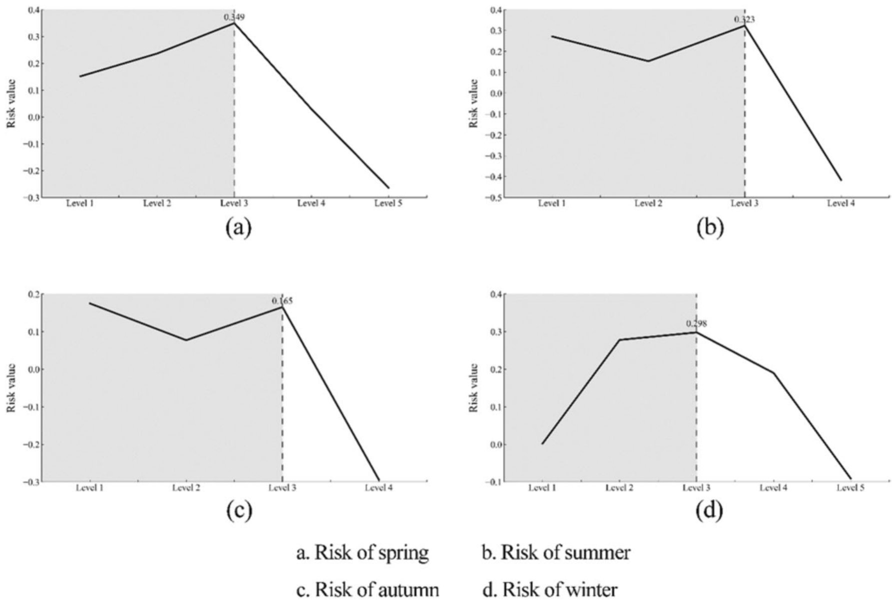


Fig. 6 Risk Area Detection Results for the Area Factor

of the two main factors significantly increases the explanation of the spatial variation in HCI. The interaction between Area and LCZtype (Area ∩ LCZtype) exhibited q-values exceeding 0.5 in all seasons (summer: 0.643; spring: 0.596; autumn: 0.603; winter: 0.516), surpassing the individual effects of Area (max 0.323) or LCZtype (max 0.179). This indicates a synergistic effect where optimal cooling is achieved through specific combinations of patch area and LCZ type. The interaction q-value was highest in summer and lowest in winter, aligning with the seasonal trend of cooling intensity. This suggests that summer heat stress amplifies the reliance of cooling effect on both patch size and urban morphology, whereas winter cooling is less dependent on their interaction due to lower solar radiation and evapotranspiration.

In the risk detection results (Fig. 6), a higher risk in the subregions indicated higher corresponding HCI values. Because of the optimal discretization of the Area factor before the geodetector analysis, the number of levels into which the Area was divided differed across seasons. Higher levels correspond to higher attribute values. The risk detection results for Area showed that the risk values generally increased from Level 1 to Level 3 and then decreased after Level 3 across the seasons. The peak at Level 3

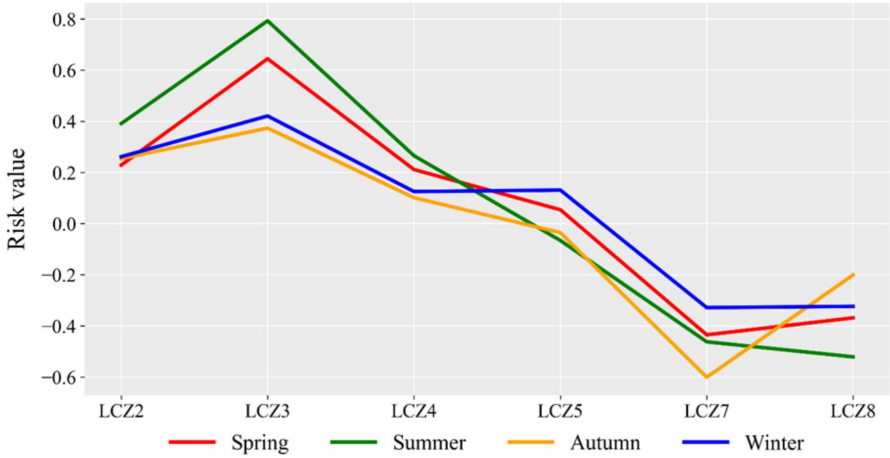


Fig. 7 Risk Area Detection Results for Local Climate Zones (LCZs)

indicates an optimal average patch area threshold of 35–50 m^2 for maximizing cooling intensity, beyond which the "edge effect" diminishes.

The risk detection results for LCZ types (Fig. 7) showed the following risk order:

Spring: LCZ3 > LCZ2 > LCZ4 > LCZ5 > LCZ8 > LCZ7
Summer: LCZ3 > LCZ2 > LCZ4 > LCZ5 > LCZ7 > LCZ8
Autumn: LCZ3 > LCZ2 > LCZ4 > LCZ5 > LCZ8 > LCZ7
Winter: LCZ3 > LCZ2 > LCZ5 > LCZ4 > LCZ8 > LCZ7

Overall, LCZ3 had the highest risk, followed by LCZ2 and LCZ4.

Discussion

Insights into Green Roof Cooling Mechanisms and Optimization Strategies

The HCIG model quantifies the temperature difference between green roofs and their surrounding environment, confirming their surface cooling effect, which is consistent with previous studies documenting green roofs' capacity to mitigate surface urban heat islands(Gong & Luo, 2019; Li et al., 2014; Wu et al., 2015). Summer was identified as the season with the best cooling effect owing to high solar radiation, which promotes plant evapotranspiration and increases cooling intensity (Jim, 2012). In summer, the highest cooling intensity in LCZ3 averaged 0.63 °C, demonstrating a significant mitigation of the heat environment even in extreme high-temperature seasons. By studying the relationship between HCI, LCZ type, and green roof landscape pattern factors, Two key factors affecting the cooling mechanism of green roofs were identified.

First, compact building areas (LCZ2, LCZ3) are highly suitable for green roofs, as their cooling intensity is significantly higher than that of open building areas (LCZ4 and LCZ5). This is likely due to the high building density and underlying surface of compact building areathe high building density and specific underlying surface characteristics of compact areas, where building density is positively correlated with the UHI effect(Chen et al., 2021). Open building areas generally have more ground-level greenery that lowers surrounding temperatures, making green-roof cooling effects more pronounced in compact contexts. Within compact building areas, LCZ3 showed better cooling effects than LCZ2, indicating that lower building heights enhanced the influence of roof cooling on the surrounding environment (Jiang et al., 2018). For urban planners, this suggests prioritizing green roof installation in residential or mixed-use compact low-rise zones (e.g., older urban districts in Guangzhou) to maximize surface temperature reduction.

Second, the average green roof area is positively correlated with cooling intensity. Green roofs provide shade by reflecting and absorbing solar radiation, and larger areas improve these shading effects. The evapotranspiration of plant leaves is also positively correlated with leaf area (Susorova et al., 2013). Geodetector analysis revealed that at Levels 1 to 3, larger average green roof areas corresponded to greater cooling intensity. Most areas with average green roof areas larger than Level 3 are open building areas, where ground-level greenery lowers the surrounding temperature more than the roof surface temperature, resulting in less significant cooling effects despite the larger green roof areas. The findings have implications for green roof design across urban contexts. Prioritize 50–150 m^2 high-cohesion patches in compact low-rise zones (LCZ3) to enhance summer cooling effects, while open high-rise areas (LCZ4) require supplementary heat-reduction measures like reflective materials.Policymakers could further support these strategies by integrating LCZ-specific green roof standards into urban planning codes.

Limitations, Future Directions, and Generalizability

Ground-level greenery significantly reduces environmental temperature (Guidelines & for Vertical Greening, Technical Guidelines for Vertical Greening., 2014), but its influence was not isolated in this study, potentially underestimating rooftop cooling effects in areas with dense ground vegetation. Future research should refine the model by excluding areas with high ground vegetation coverage or by separating the contributions of ground and rooftop greenery using combined remote sensing and field measurements. Additionally, LCZ classifications did not distinguish between commercial and residential areas. In future research, the cooling effects of rooftop greening could be separately assessed for commercial and residential areas, providing guidance for implementing targeted, zone-specific green roof strategies to further optimize urban thermal environment management. Separating these zones could enable targeted green roof strategies for distinct urban functions.

The findings of this study can provide valuable reference for other regions with similar climatic conditions and urban morphology. In particular, in subtropical monsoon climates and high-density urban settings, the cooling effect of rooftop greening can be expected to be transferable. In urban planning, the layout and scale of rooftop greening can be optimized according to the characteristics of different local climate zones, enabling more efficient cooling and a more rational spatial configuration, thereby providing scientific guidance for sustainable urban development and low-carbon transformation.

Conclusions

Green roofs can effectively address the growing scarcity of urban construction land and the decreasing availability of green space by making full use of the idle space on building rooftops. Through shading and evapotranspiration, green roofs help mitigate the increasingly severe UHI effect. Urban heat islands can be classified based on their spatial occurrence into boundary layer, canopy, surface, and subsurface UHIs. This study focused on surface UHIs and explored the mechanisms by which green roofs mitigate the UHI effect across different seasons by considering LCZ types and the intrinsic properties of green roofs. The following conclusions were drawn:

1. Green roofs can significantly reduce surface temperatures in urban areas, thereby mitigating the surface UHI effect. Among the four seasons, summer is when green roofs exhibit the best cooling effect, with the highest average cooling intensity reaching 0.63 °C.
2. The LCZ3 type, characterized by compact buildings, is the most effective local climate zone for green roofs in terms of cooling. Compact building areas are more suitable for green roofs than are open building areas. The larger the average area of the green roofs, the better is the cooling effect.

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Data Availability Data will be made available from the corresponding author on reasonable request.

Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

References

Santamouris, M. (2015). Regulating the damaged thermostat of the cities—Status, impacts and mitigation challenges. *Energy and Buildings, 91*, 43–56. https://doi.org/10.1016/j.enbuild.2015.01.027

Qin, D., & Stocker, T. (2014). Highlights of the IPCC fifth assessment report, working Group I. *Clim Change Res Prog, 10*, 1–6. https://doi.org/10.3969/j.issn.1673-1719.2014.01.001

Zhong-Wei, Y., Jun, W., Jiang-Jiang, X., & Jin-Ming, F. (2016). Review of recent studies of the climatic effects of urbanization in China. *Climate Change Research Progress: English Edition, 7*, 15. https://doi.org/10.1016/j.accre.2016.09.003

Shi, T., Huang, Y., Wang, H., Shi, C. E., & Yang, Y. J. (2015). Influence of urbanization on the thermal environment of meteorological station: Satellite-observed evidence. *Advances in Climate Change Research, 6*, 7–15. https://doi.org/10.1016/j.accre.2015.07.001

Peng, B., Shi, Y., Wang, H., Wang, Y. (2013). The influencing mechanism and action law of urban heat island effect—A case study of Shanghai. Acta Geogr Sin, 68, 11. https://doi.org/10.11821/dlxb201311002

Zhang, N., Gao, Z., Wang, X., & Chen, Y. (2010). Modeling the impact of urbanization on the local and regional climate in Yangtze River Delta, China. *Theoretical and Applied Climatology, 102*, 331–342. https://doi.org/10.1007/s00704-010-0263-1

Huang, Q., & Lu, Y. (2017). Urban heat island research from 1991 to 2015: A bibliometric analysis. *Theoretical and Applied Climatology*. https://doi.org/10.1007/s00704-016-2025-1

Kovats, R. S., & Hajat, S. (2008). Heat stress and public health: A critical review. *Annual Review of Public Health, 29*, 41–55. https://doi.org/10.1146/annurev.publhealth.29.020907.090843

Hou, A., Ni, G., Yang, H., & Lei, Z. (2013). Numerical analysis on the contribution of urbanization to wind stilling: An example over the greater Beijing metropolitan area. *Journal of Applied Meteorology and Climatology, 52*, 1105–1115. https://doi.org/10.1175/JAMC-D-12-013.1

Amir, B., Sailor, D.J., Crank, P.J., Ban-Weiss, G.A. (2018). Direct and indirect effects of high-albedo roofs on energy consumption and thermal comfort of residential buildings. Energy Build, 178, S0378778818321431-. https://doi.org/10.1016/j.enbuild.2018.08.048.

Rizwan, A. M., Dennis, L. Y. C., & Liu, C. (2008). A review on the generation, determination and mitigation of urban heat island. *Journal of Environmental Sciences, 20*, 120–128. https://doi.org/10.1016/S1001-0742(08)60019-4

Gong, X., Luo, T. (2019). Research progress of green roofs in mitigating urban heat island effect—A review of core journals in Web of Science from 2003 to 2018. In Proceedings of the 2019 Annual Conference of the Chinese Society of Landscape Architecture.

Wu, Z., Zou, M., Chen, X., Ai, L., Xian, X. (2015). Cooling and Humidifying Effects of Rooftop Greening in Chongqing, Vol. 42, Fujian Forestry Science and Technology, pp. 87–91 + 99. https://doi.org/10.13428/j.cnki.fjlk.2015.01.019.

Li, D., Bou-Zeid, E., & Oppenheimer, M. (2014). The effectiveness of cool and green roofs as urban heat island mitigation strategies. *Environmental Research Letters, 9*, Article 055002. https://doi.org/10.1088/1748-9326/9/5/055002

Jamei, E., Thirunavukkarasu, G., Chau, H. W., Seyedmahmoudian, M., Stojcevski, A., & Mekhilef, S. (2023). Investigating the cooling effect of a green roof in Melbourne. *Building And Environment*. https://doi.org/10.1016/j.buildenv.2023.110965

Zheng, X., Kong, F., Yin, H., Middel, A., Yang, S., Liu, H., & Huang, J. (2023). Green roof cooling and carbon mitigation benefits in a subtropical city. *Urban for Urban Greening, 86*, Article 128018. https://doi.org/10.1016/j.ufug.2023.128018

Feng, Y., Wang, J., Zhou, W., Li, X., & Yu, X. (2022). Evaluating the cooling performance of green roofs under extreme heat conditions. *Frontiers in Environmental Science, 10*, Article 874614. https://doi.org/10.3389/fenvs.2022.874614

Gao, L., Zhang, L., Ye, M., Guo, L., & Zhou, Z. (2021). Analysis of the urban heat island effect in Guangzhou from 2013 to 2018 based on remote sensing. *Environment and Ecology, 3*, Article 7.

Stewart, I. D., & Oke, T. R. (2012). Local climate zones for urban temperature studies. *Bulletin of the American Meteorological Society, 93*, 1879–1900. https://doi.org/10.1175/BAMS-D-11-00019.1

Zheng, Y., Ren, C., Xu, Y., Wang, R., Ho, J., Lau, K., & Ng, E. (2018). GIS-based mapping of Local Climate Zone in the high-density city of Hong Kong. *Urban Clim, S2212095517300445*(24), 419–448. https://doi.org/10.1016/j.uclim.2017.05.008

Benjamin, B., Paul, A., Jürgen, B.h., Jason, C., Olaf, C., Johannes, F., Gerald, M., Linda, S., Iain, S. (2015). Mapping local climate zones for a worldwide database of the form and function of cities. Int J Geo Inf, 4, 199–219. https://doi.org/10.3390/ijgi4010199

Zhao, X., Liu, X., Gao, K. (2017). Surface temperature Inversion based on Radiation Conduction Equation. Beijing Surv Mapp, 5. https://doi.org/10.19580/j.cnki.1007-3000.2017.03.021

Raissouni, N., & Sobrino, J. A. (2000). Toward remote sensing methods for land cover dynamic monitoring: Application to Morocco. *International Journal of Remote Sensing, 21*, 353–366. https://doi.org/10.1080/014311600210876

Qin, Z., Li, W., Xu, B., Chen, Z., Liu, J. (2004). Estimation of surface emissivity in the range of landsat TM6 band. Remote Sens Land Resour, 3. https://doi.org/10.3969/j.issn.1671-6647.2004.z1.022

Jiang, Z., Peng, L., Yang, X., Yao, L., & Zhu, C. (2018). Thermal effects of green roofs at the neighborhood scale and their relationship with urban morphological structures. *Acta Ecologica Sinica, 38*, 7120–7134. https://doi.org/10.5846/stxb201708111449

Wang, J., Xu, C. (2017). Geodetector: Principles and prospects. Acta Geogr Sin, 72. https://doi.org/10.11821/dlxb201701010

Jim, C. Y. (2012). Effect of vegetation biomass structure on thermal performance of tropical green roof. *Landscape and Ecological Engineering, 8*, 173–187. https://doi.org/10.1007/s11355-011-0161-4

Chen, G., Li, N., Cai, Y., He, M. (2021). Analysis of Summer Heat Island Intensity in Guangzhou Based on Local Climate Zones, Vol. 37, Architectural Press Science, p. 9. https://doi.org/10.13614/j.cnki.11-1962/tu.2021.06.13

Susorova, I., Angulo, M., Bahrami, P., & Stephens, B. (2013). A model of vegetated exterior facades for evaluation of wall thermal performance. *Building and Environment, 67*, 1–13. https://doi.org/10.1016/j.buildenv.2013.04.027

Shanghai Municipal Greenery and Public Sanitation Bureau. (2014). Technical Guidelines for Vertical Greening, Technical Guidelines for Vertical Greening.